Problem Set 4

Due Wednesday, Dec 18, 2024

Optimal Receptive Field in 1-Dimensional World

We will follow our lecture on the efficient coding principle, which allows us to derive the optimal synaptic weight W (also called receptive field for the output neurons) that maximizes the mutual information in the presence of noise. Go through page 5-9 in the lecture note carefully. Now imagine a 1-dimensional visual world on a circle with inputs that are correlated according to

$$c(n.m) = \langle x(n)x(m) \rangle = c(n-m) = \exp(-\frac{|n-m|}{D}), \tag{1}$$

where n is a spatial index running from 1 to N (note: with periodic boundary conditions, $|n-m|$ for distances larger than $N/2$ should be replaced by $N - |n-m|$). The input contains additional spatially uncorrelated input noise with variance σ_z^2. We want to find the shape of the receptive fields, namely the functional form of W that are optimized for this world.

(a) Using my lecture note as a guide, show that in the spatial frequency domain, the optimal receptive field (or filter) $w(k)$ is given by the equation

$$w^2(k) = \frac{\sigma_\eta^2}{\sigma_z^2}\left(\frac{1}{2}\left[\frac{c(k)}{c(k)+\sigma_z^2}\left(1+\sqrt{1+\frac{\mu}{c(k)}}\right)\right]-1\right),$$

where $c(k)$ is power spectrum of the visual inputs, σ_η^2 is the output noise. We will set $\sigma_\eta^2 = 1$ in this model. Please recall my lectures what is a power spectrum and how it is related to the eigenvalues of the input covariance matrix, which happens to be a circulant matrix in our 1-dimensional visual world. μ is a term involving the lagrange multiplier λ, more precisely $\mu \sim 1/\lambda$. Here, we make a specific requirement that the total variance (or power) of the outputs $\mathbf{y}$ is independent of input noise, and is set as a constant.

(b) Find the dependence of $w(k)$ on $c(k)$ analytically, in the limits of small and large input noise σ_z^2. Interpret your results. **Guide**: In order to derive these limits, you will need to consider how μ varies with σ_z^2 so that the total output variance remains a constant. It can be shown that for small noise $\mu \propto \sigma_z^2$, where as for large noise, $\mu \propto \sigma_z^4$.

(c) Numerically Fourier transform $W(k)$ to calculate the spatial shape of the optimal filters. Use $N = 401$ and $D = 30$ and study different values of noise, σ_z^2. Qualitatively explain your results, by plotting the spectrum $W(k)$ vs. k and $W(n)$ vs. n for different values of input noise. Compare these numerical results with your analytical predictions from (b). You will find the matlab functions fft and ifft useful.